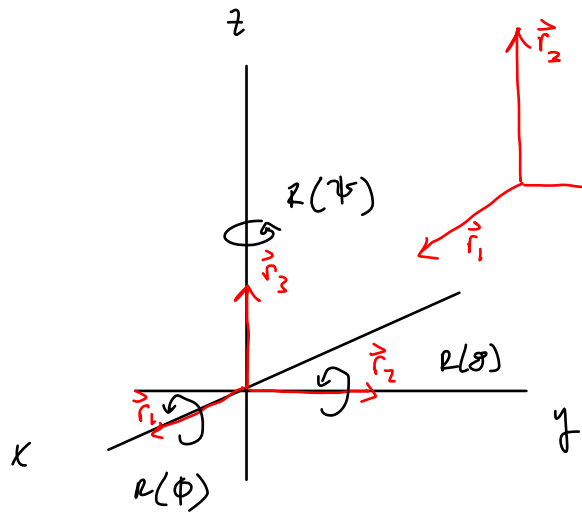


$$r_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$r_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$r_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$R_{r_1}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix}$$

$$R_{r_2}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

$$R_{r_3}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$